

Supporting Information for “What Must a Pore-Scale Transport Model Remember? Memory Adequacy in Non-Fickian Transport”

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Purpose and Reproducibility

This supporting information records the numerical details behind the relative memory-adequacy results reported in the main text. The main manuscript keeps the scientific design compact: a resolved trajectory archive, competing transition mechanisms, and held-out observables. The tables and figures below provide the audit trail for that design, including proxy-observable sensitivity, geometric-support diagnostics, archive-density and segment-length diagnostics, the tight OpenFOAM resolution ladder, particle-tracking tolerances, objective-weight sensitivity, and the evidence matrix used to construct the relative retained-state map.

The OpenFOAM cases use voxel finite-volume meshes extracted from the same Core2 Benheimer subvolume. The 18 μm and 12 μm cases use downsampled voxel meshes, while the 6 μm case uses the full image-resolution voxel mesh and a strict SIMPLE solve with zero linear-solver relative tolerances and residual-control thresholds of 10^{-9} for pressure and velocity. All manuscript-facing trajectory archives contain 20,000 particles. The OpenFOAM archives use 4000 saved steps and saved-step interval $\Delta t = 0.1$. Particle integration used internal substepping with physical tolerances scaled across the resolution ladder; no case reached the substep cap.

Mechanisms with the lowest mean objective in these tables should not be read as statistically separated closures. Several mean differences are small compared with split-to-split variability, and separate columns report mean rank and split wins where available. The tables support qualitative claims: breakthrough and the occupied-volume proxy can disagree, lowest-mean mechanisms and mixture weights shift with objective and regime, OpenFOAM preserves velocity autocorrelation under refinement, and no retained state wins universally.

The occupied-volume and encounter quantities are proxy observables. The dilution index used here is an entropy-based occupied-volume measure computed from binned particles, not a resolved concentration field or scalar-dissipation calculation. Encounter probability is a coarse-grained encounter-opportunity proxy for sampled particle pairs, not a reactive-transport rate law. We therefore report reference-split, bin-size, encounter-radius, inlet-neighborhood pair-sampling, archive-density, segment-length, and geometric-support sensitivity checks in Figures S1–S2 and Tables S1–S4. These checks support qualitative interpretation of the memory map, not calibrated reactive-transport prediction.

Solver and Tracking Audit

Supporting Tables

Proxy-observable sensitivity checks

Core1 baseline, 20,000-particle archive; generated ensembles are re-evaluated under metric perturbations.

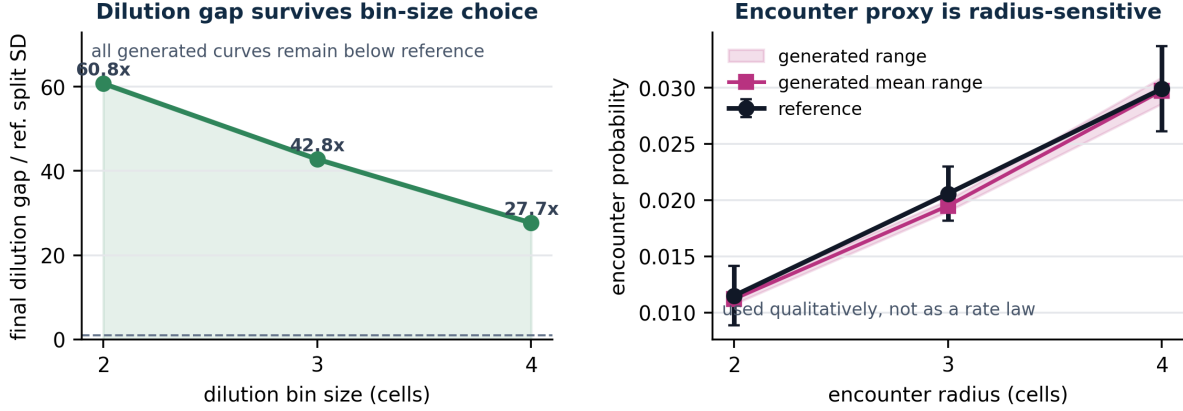


Figure S1: Core1 baseline proxy-observable sensitivity checks. The occupied-volume panel varies the entropy-bin size and reports the final-time generated-reference gap normalized by reference split-to-split variability. The encounter-opportunity panel varies the proximity radius and compares the reference probability with the generated range across candidate mechanisms.

Table S1: Core1 baseline sensitivity checks for proxy observables. Reference-vs-reference errors are computed by splitting each held-out reference test set in half. Occupied-volume-bin rows report the closest generated final-time occupied-volume value among candidate mechanisms; encounter-radius rows report the mechanism with the lowest mean absolute encounter-opportunity proxy error.

Check	Perturbation	Reference result	Generated result	Qualitative effect
Reference split	default metrics	BTC score 7.30 ± 1.82 ; occupied-volume log MAE 0.017 ± 0.006 ; pair MAE 0.68 ± 0.20 ; encounter-opportunity abs. error 0.0041 ± 0.0015	Defines empirical uncertainty scale	Used to interpret Figure 3 ribbons and the occupied-volume gap inset
Occupied-volume bin	2 cells	final occupied volume 9838 ± 146	closest generated 946; gap 60.8 ref. SD	all generated curves remain below reference
Occupied-volume bin	3 cells	final occupied volume 17424 ± 334	closest generated 3121; gap 42.8 ref. SD	manuscript result unchanged
Occupied-volume bin	4 cells	final occupied volume 25359 ± 667	closest generated 6910; gap 27.7 ref. SD	occupied-volume gap persists under coarser bins
Encounter radius	2 cells	probability 0.0115 ± 0.0026	generated range 0.0108–0.0117; lowest error bootstrap mixture	encounter-opportunity ranking is radius-sensitive
Encounter radius	3 cells	probability 0.0206 ± 0.0024	generated range 0.0190–0.0200; lowest error Gaussian/Bayes	default proxy remains qualitative
Encounter radius	4 cells	probability 0.0299 ± 0.0038	generated range 0.0286–0.0309; lowest error Gaussian/Bayes	proximity metric is not a reaction-rate law

Table S2: Core1 baseline sensitivity of pair-separation and encounter-opportunity proxies to pair-sampling convention. Uniform sampling draws particle pairs from all ensemble indices, as in the main benchmarks. The inlet-neighborhood diagnostic instead samples pairs whose initial transverse separation is within 6 grid cells. The conditioned diagnostic changes row-level lowest-error mechanisms, so it supports the interpretation of pair and encounter metrics as ensemble-level proximity diagnostics rather than labeled-reactant pair histories.

Pair sampling	Reference encounter probability	Lowest pair-separation error	Lowest encounter-opportunity error	Interpretation
Uniform over all indices	0.0206	hybrid; pair MAE 1.459	Gaussian/Bayes; abs. error 0.0036	Main benchmark convention
Initial transverse separation ≤ 6 cells	0.4624	pair-rerank; pair MAE 0.536	hybrid; abs. error 0.0113	Row-level lowest-error mechanisms depend on the pair-sampling rule

Archive-density and segment-length diagnostic

Core1 baseline Gaussian/Bayes recombination; 180 generated paths per repeat, 3 repeats per setting.

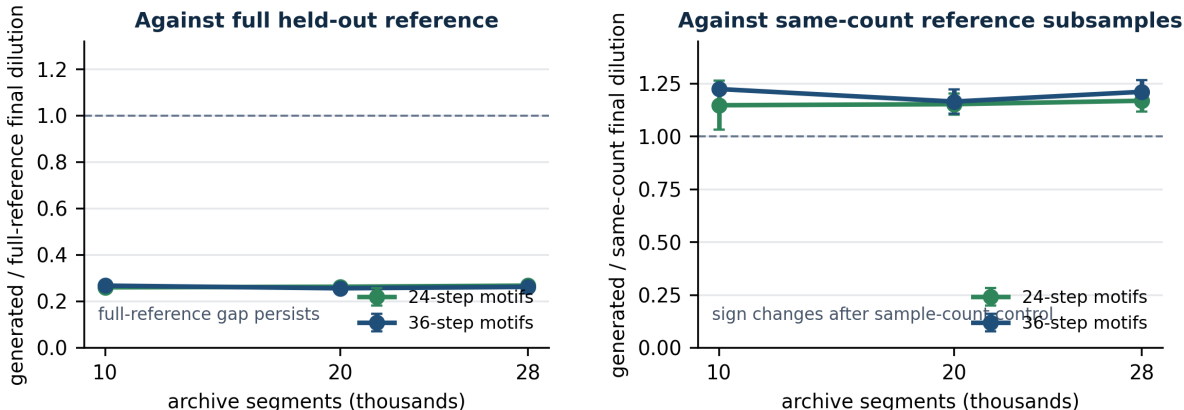


Figure S2: Limited Core1 archive-density and segment-length convergence diagnostic for the occupied-volume gap. The diagnostic uses Gaussian/Bayes recombination with 10k, 20k, and 28k archived segments, corresponding to 5k, 10k, and 14k training trajectories after the 30% held-out split. Two motif lengths are tested. Comparing recombined ensembles with the full held-out reference preserves the apparent occupied-volume deficit, but comparing with same-count held-out reference subsamples changes the sign. This indicates that the Figure 3 occupied-volume contrast is a proxy-observable diagnostic affected by entropy-estimator sample support, not a calibrated universal dilution deficit.

Table S3: Core1 archive-density and segment-length diagnostic. Values are means over three repeats. Ratios are final-time entropy-based occupied-volume indices for recombined trajectories divided by either the full held-out reference or a same-count held-out reference subsample.

Motif	Archive	Segments	BTC score	Occ.-vol. log MAE	Gen/full ref.	Gen/same-count ref.
24-step	5k particles	10k	140.65	1.096	0.261	1.148
24-step	10k particles	20k	124.84	1.091	0.263	1.153
24-step	14k particles	28k	133.01	1.082	0.268	1.169
36-step	5k particles	10k	119.25	1.071	0.268	1.225
36-step	10k particles	20k	108.17	1.105	0.256	1.165
36-step	14k particles	28k	108.28	1.064	0.262	1.212

Table S4: Core1 baseline geometric-support diagnostic for recombined trajectories. Violation fractions are computed by flooring each recombined point to a voxel and checking the connected pore mask used by the resolved tracker. Occupied-volume errors are mean held-out occupied-volume log MAE values from the 20,000-particle outer-split benchmark. The held-out resolved reference sample has 0% violation under the same check. Across the six recombined mechanisms, violation rate is not positively correlated with occupied-volume error (Pearson $r = -0.50$, Spearman $r = -0.43$), so out-of-pore recombination is interpreted as a broad support limitation rather than a mechanism-specific explanation for the occupied-volume ranking.

Mechanism	Violating points	In-solid points	Out-of-domain points	Occ.-vol. log MAE
Bootstrap mixture	52.63%	47.63%	5.00%	1.470 ± 0.050
Validation mixture	52.55%	47.46%	5.08%	1.456 ± 0.044
Gaussian/Bayes	52.16%	46.49%	5.67%	1.450 ± 0.048
kNN conditional	50.89%	46.53%	4.37%	1.457 ± 0.031
hybrid	54.47%	48.61%	5.87%	1.455 ± 0.049
pair-rerank	50.68%	45.72%	4.96%	1.487 ± 0.052
Held-out reference	0.00%	0.00%	0.00%	—

Table S5: Archive and validation bookkeeping for manuscript-facing benchmarks. All rows use 20,000 tracked particles before splitting, a 70/30 outer train–test split, four outer splits, and three inner validation repeats. Candidate caps are listed as memory-free / archive-proximity / velocity-continuity, learned-context, and pair-organization candidates per transition; mixtures combine the component transition distributions.

Benchmark group	Outer train/test	Inner fit/validation	Generated trajectories	Segment / match / stride	Archived motifs	Candidate motifs	Pair samples
Graph-flow balanced outer-split benchmarks (Core1 Pe sweep; Core2 graph)	14,000 / 6,000	9,800 / 4,200	60 per validation-grid candidate; 120 per test mechanism per outer split	36 / 20 / 400	19,600 inner fit; 28,000 final train	all / 96 / 256	3,000
Graph-flow objective-sensitivity and breakthrough-only counterfactual	14,000 / 6,000	9,800 / 4,200	50 per validation-grid candidate; 100 per test mechanism or validation mixture per outer split	36 / 20 / 400	19,600 inner fit; 28,000 final train	all / 96 / 256	2,500
OpenFOAM balanced resolution ladder	14,000 / 6,000	9,800 / 4,200	60 per validation-grid candidate; 120 per test mechanism per outer split	160 / 80 / 1600 at 18 μm ; 160 / 60 / 1600 at 12 and 6 μm	29,400 inner fit; 42,000 final train	all / 96 / 256	3,000
OpenFOAM objective-sensitivity ladder	14,000 / 6,000	9,800 / 4,200	50 per validation-grid candidate; 100 per test mechanism or validation mixture per outer split	160 / 80 / 1600 at 18 μm ; 160 / 60 / 1600 at 12 and 6 μm	29,400 inner fit; 42,000 final train	all / 96 / 256	2,500
Core1 particle-count stability diagnostic	Subsets of 500, 2,500, 5,000, 10,000, and 20,000 compared with 20,000-particle reference	No archive recombination	30 random subset repeats except the full reference	Not applicable	Not applicable	Not applicable	10,000 per subset; 50,000 reference

Table S6: OpenFOAM resolution-ladder and particle-tracking audit for Core2. The 6 μm case is the strict full-resolution solve; the listed residual is the largest final OpenFOAM initial residual among pressure and velocity components at convergence. Lower-resolution residuals were not rerun with the same strict residual audit, so the table emphasizes their accepted flow fields and particle-integration diagnostics.

Voxel size	Cells	Flow time	k (m^2)	Max residual	Particles	Max substeps	Cap hits
18 μm	98,270	103	4.277×10^{-12}	—	20,000	4	0
12 μm	322,524	105	3.690×10^{-12}	—	20,000	5	0
6 μm	2,650,688	604	3.492×10^{-12}	9.94×10^{-10}	20,000	10	0

Table S7: Tight particle-tracking tolerances across the OpenFOAM resolution ladder. The tolerance scaling preserves physical distance rather than voxel count. Mean substeps are reported per saved output step.

Voxel size	D	Max adv. step	Max diff. step	Mean length	Mean substeps
18 μm	10^{-3}	0.0333	0.0250	4000.998	1.0082
12 μm	2.25×10^{-3}	0.0500	0.0375	4000.936	1.0106
6 μm	9×10^{-3}	0.1000	0.0750	4000.805	1.0094

Table S8: Outer-split variability on 20,000-particle Core1 baseline trajectories. Lower objective and rank are better; mean differences should be interpreted against the split-to-split standard deviations.

Mechanism	Mean obj.	Std. obj.	Mean rank	Split wins	Beats GB/H
Gaussian/Bayes	319.51	8.35	3.00	0	-/2
kNN conditional	319.74	22.51	2.50	2	3/2
hybrid	321.94	14.63	2.50	1	2/-
Bootstrap mixture	325.96	12.44	3.50	1	1/1
Validation mixture	327.99	12.69	4.25	0	2/1
pair-rerank	345.03	22.97	5.25	0	0/0

Table S9: Objective-sensitivity lowest-mean mechanism under the eight Core1 scalar objective-weight regimes used in Figure 4. The regimes perturb the objective to reveal observable dependence rather than to define a universal scalar score.

Regime	Lowest-mean mechanism	Mean obj.	Mean rank
Balanced	Gaussian/Bayes	343.69	3.50
Breakthrough only	Bootstrap mixture	102.30	2.25
BTC heavy	Gaussian/Bayes	439.82	2.25
Pair heavy	Bootstrap mixture	290.50	1.75
Occupied-volume heavy	Gaussian/Bayes	655.01	2.50
Encounter light	Gaussian/Bayes	339.91	3.25
Encounter heavy	Bootstrap mixture	177.99	2.75
No encounter	Gaussian/Bayes	339.49	3.25

Table S10: Mean pooled-validation mixture weights across the Core1 Peclet sweep. Weights are ordered as velocity memory, archive proximity, learned context, and pair organization.

Condition	Gaussian/Bayes	kNN	hybrid	pair-rerank
$D = 3 \times 10^{-4}$	0.2292	0.3333	0.3125	0.1250
$D = 10^{-3}$	0.2500	0.2292	0.2708	0.2500
$D = 3 \times 10^{-3}$	0.3750	0.2292	0.3125	0.0833

Table S11: Core2 18 μm OpenFOAM tight-archive outer-split mechanism performance with 20,000 tracked particles. Lower objective and rank are better; the spreads are large enough that the table supports lowest-mean, mean-rank, and split-win comparisons rather than crisp statistical separation.

Mechanism	Mean obj.	Std. obj.	Mean rank	Split wins
kNN conditional	1570.01	29.49	3.75	1
hybrid	1571.85	102.93	2.75	1
pair-rerank	1578.50	98.33	3.25	0
Gaussian/Bayes	1580.43	106.41	3.75	1
Bootstrap mixture	1594.09	175.76	3.75	1
Validation mixture	1634.54	83.50	3.75	0

Note. Table S11 and Table S16 report lowest mean objective for the balanced outer-split benchmark. Table S12 reports objective-sensitivity lowest-mean mechanisms under perturbed scalar objectives. Differences at 18 μm are small relative to split variability; the interpretation is dependence on validation target, not a unique 18 μm closure.

Table S12: Objective-sensitivity lowest-mean retained state under different objective-weight regimes for the 20,000-particle OpenFOAM resolution ladder. This table reports lowest-mean relative summaries, not statistically certified unique closures: learned transition context has the lowest mean objective across most 18 μm regimes, velocity continuity has the lowest mean objective at 12 μm , and validation mixtures have the lowest mean objective in most 6 μm sensitivity tests.

Objective regime	18 μm	12 μm	6 μm
Balanced	Learned context	Velocity memory	Validation mixture
Breakthrough only	Learned context	Velocity memory	Validation mixture
BTC heavy	Learned context	Velocity memory	Validation mixture
Pair heavy	Learned context	Velocity memory	Bootstrap mixture
Occupied-volume heavy	Velocity memory	Velocity memory	Bootstrap mixture
Encounter light	Learned context	Velocity memory	Validation mixture
Encounter heavy	Learned context	Velocity memory	Validation mixture
No encounter	Learned context	Velocity memory	Validation mixture

Note. Table S11 and Table S16 report lowest mean objective for the balanced outer-split benchmark. Table S12 reports objective-sensitivity lowest-mean mechanisms under perturbed scalar objectives. Differences at 18 μm are small relative to split variability; the interpretation is dependence on validation target, not a unique 18 μm closure.

Table S13: Memory hypotheses evaluated in the main manuscript.

Memory hypothesis	Implementation	Primary information used	Role
Memory-free	Unconditional	Archive frequency only	Null conditioning
Archive proximity	kNN conditional	Local empirical proximity	Nonparametric matching
Velocity memory	Gaussian/Bayes	Diffusion-scaled velocity continuity	Original physics kernel
Learned context	hybrid	Contrastive learned score plus Gaussian/Bayes	Learned transition memory
Pair organization	pair-rerank	Pair descriptors plus Gaussian/Bayes	Short-horizon spatial organization
Memory mixture	Validation mixture	Component weights chosen by held-out score	Objective-aware memory weighting

Table S14: Trajectory ensembles and benchmark roles. All manuscript-facing ensembles contain 20,000 particles. In the OpenFOAM resolution ladder, $u_{\text{grid}} \propto \Delta x^{-1}$ and $D_{\text{grid}} \propto \Delta x^{-2}$ so that physical speed, physical diffusivity, and Peclet number based on the 18 μm reference length are held approximately fixed.

Ensemble	Geometry	Velocity field	Diffusivity	Role
Core1 baseline	Core1	Graph-Laplace	10^{-3}	Baseline outer-split mixture benchmark
Core1 high Pe	Core1	Graph-Laplace	3×10^{-4}	Peclet-regime sensitivity
Core1 low Pe	Core1	Graph-Laplace	3×10^{-3}	Peclet-regime sensitivity
Core2 graph	Core2	Graph-Laplace	10^{-3}	Geometry transfer
Core2 OpenFOAM 18 μm	Core2	Finite volume	10^{-3}	Tight flow-fidelity test
Core2 OpenFOAM 12 μm	Core2	Finite volume	2.25×10^{-3}	Tight resolution-ladder test
Core2 OpenFOAM 6 μm	Core2	Finite volume	9×10^{-3}	Strict full-resolution flow test

Table S15: Evidence matrix across the benchmark conditions. Weights are ordered as velocity continuity, archive proximity, learned context, and pair organization. The “lowest-mean memory” column denotes the mechanism with the lowest mean objective under the stated benchmark, not a statistically certified unique or universal closure.

Condition	Lowest-mean memory	Mean obj.	Mean rank	Mixture weights
Core1 high Pe	Learned context	316.35	2.00	0.229, 0.333, 0.313, 0.125
Core1 baseline	Velocity memory	319.51	3.00	0.250, 0.229, 0.271, 0.250
Core1 low Pe	Validation mixture	319.34	1.50	0.375, 0.229, 0.313, 0.083
Core2 graph	Validation mixture	325.98	3.00	0.271, 0.125, 0.396, 0.208
Core2 OpenFOAM 18 μm	Archive proximity	1570.01	3.75	0.083, 0.396, 0.250, 0.271
Core2 OpenFOAM 12 μm	Velocity memory	1647.69	2.00	0.125, 0.500, 0.125, 0.250
Core2 OpenFOAM 6 μm	Pair organization	1657.02	2.50	0.313, 0.396, 0.083, 0.208

Note. Table S11 and Table S16 report lowest mean objective for the balanced outer-split benchmark. Table S12 reports objective-sensitivity lowest-mean mechanisms under perturbed scalar objectives. Differences at 18 μm are small relative to split variability; the interpretation is dependence on validation target, not a unique 18 μm closure.

Table S16: OpenFOAM resolution ladder for Core2 using tight 20,000-particle trajectory archives. The autocorrelation values are axial velocity autocorrelations of the resolved reference trajectories at the listed saved-step lag with $\Delta t = 0.1$. The near equality between 12 and 6 μm shows that velocity persistence is stable over the two finer meshes; the permeability values tighten substantially but should not be read as a three-point asymptotic convergence proof.

Voxel size	Cells	k (m^2)	Corr. lag 10	Corr. lag 80	Balanced lowest mean	Mean obj.
18 μm	98,270	4.277×10^{-12}	0.108	0.086	Archive proximity	1570.01
12 μm	322,524	3.690×10^{-12}	0.124	0.101	Velocity memory	1647.69
6 μm	2,650,688	3.492×10^{-12}	0.120	0.100	Pair organization	1657.02

Note. Table S11 and Table S16 report lowest mean objective for the balanced outer-split benchmark. Table S12 reports objective-sensitivity lowest-mean mechanisms under perturbed scalar objectives. Differences at 18 μm are small relative to split variability; the interpretation is dependence on validation target, not a unique 18 μm closure.